# Removing second countability from Naderi's reduced Fourier–Stieltjes weak* fixed point theorem

arXiv automatic math run `fa_banach_001`

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## Source and target

The source paper is Fouad Naderi, *C*-algebraic approach to fixed point theory generalizes Baggett's theorem to groups with discrete reduced duals*, arXiv:1612.08286, later published in J. Math. Anal. Appl. 461 (2018), 451–460 [1]. Theorem 3.1 of the source proves:

> If the reduced Fourier–Stieltjes algebra $B_\rho(G) = C_r^*(G)^*$ of a second countable locally compact group $G$ has the weak* fixed point property, then $G$ is compact.

On page 9, the paper asks whether the second countability hypothesis can be removed:

> We would like to conclude our work with the following questions:
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> **Question 1.** Is there any counterexample of a non-expansive action to BGC?
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> **Question 2.** Is BGC true for amenable semigroups?
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> The norm of C*-algebras are completely determined by the algebraic properties of the space. Inspired by Theorem 4.1, we ask:
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> **Question 3.** Is BGC true for non-unital C*-algebras?
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> From Theorem 4.1, we know the preduals of von Neumann algebras enjoy BGC property. Now, we ask:
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> **Question 4.** Can we characterize all Banach spaces with BGC property?
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> **Question 5.** Can one prove Theorem 3.1 for weak* fpp without using the second countability assumption on $G$?


> **Question 5.** Can one prove Theorem 3.1 for weak* fpp without using the second countability assumption on $G$?


## Result

**Theorem 1.** *Let $G$ be an arbitrary locally compact group. If $B_\rho(G) = C_r^*(G)^*$, with its canonical $C_r^*(G)$-weak* topology, has the weak* fixed point property for nonexpansive mappings, then $G$ is compact.*

Together with Naderi's compact-group converse [1, Proposition 3.3], this gives the countability-free equivalence:

$$G \text{ compact} \quad \Longleftrightarrow \quad B_\rho(G) \text{ has weak* fpp.}$$

In fact the compact direction gives the stronger weak* fpp for left reversible semigroups.

## Proof Intuition

The second countability in Naderi's proof is used to make $C_r^*(G)$ separable and then invoke the separable C*-algebra machinery. Instead of trying to make the whole group separable, pass to a separable quotient where noncompactness is still visible.

Every noncompact locally compact group has a noncompact $\sigma$-compact open subgroup $H$. By the Kakutani–Kodaira theorem, $H$ has a compact normal subgroup $N$ such that $H/N$ is second countable; since $N$ is compact and $H$ is noncompact, $H/N$ is still noncompact. The weak* fixed point property of $B_\rho(G)$ passes first to $B_\rho(H)$ by the open-subgroup compression $C_r^*(G) \to C_r^*(H)$, and then to $B_\rho(H/N)$ by the quotient map $C_r^*(H) \to C_r^*(H/N)$. Naderi's second-countable theorem applied to $H/N$ then forces $H/N$ compact, a contradiction.

## Proof

We first record two elementary permanence facts.

**Lemma 1.** *Let $A$ and $B$ be Banach spaces. Suppose $J : B^* \to A^*$ is a weak*-to-weak* continuous linear isometry. If $A^*$ has the weak* fixed point property, then $B^*$ has the weak* fixed point property.*

*Proof.* Let $K \subset B^*$ be nonempty, convex, and weak* compact, and let $T : K \to K$ be nonexpansive. Then $J(K)$ is a nonempty convex weak* compact subset of $A^*$. Define $\widetilde{T} : J(K) \to J(K)$ by $\widetilde{T}(Jx) = J(Tx)$. Since $J$ is an isometry, $\widetilde{T}$ is nonexpansive. A fixed point $Jx$ of $\widetilde{T}$ gives $Tx = x$. $\square$

**Lemma 2.** *Let $G$ be a locally compact group.*

1. *If $H$ is an open subgroup of $G$, then weak* fpp of $B_\rho(G)$ implies weak* fpp of $B_\rho(H)$.*
2. *If $N$ is a compact normal subgroup of $G$, then weak* fpp of $B_\rho(G)$ implies weak* fpp of $B_\rho(G/N)$.*

*Proof.* For (1), identify $L^2(H)$ with the subspace of $L^2(G)$ consisting of functions supported on $H$, and let $P$ be the orthogonal projection from $L^2(G)$ onto this subspace. For $f \in C_c(G)$, compression gives

$$P\lambda_G(f)|_{L^2(H)} = \lambda_H(f|_H).$$

Hence compression extends by continuity to a contractive projection

$$E_H : C_r^*(G) \longrightarrow C_r^*(H)$$

which is the identity on the canonical copy of $C_r^*(H)$. Therefore $E_H^* : B_\rho(H) \to B_\rho(G)$ is a weak*-continuous isometry. Lemma 1 applies.

For (2), write $q : G \to G/N$ for the quotient map and normalize Haar measure on the compact group $N$. The averaging map

$$Q : C_c(G) \to C_c(G/N), \qquad Qf(gN) = \int_N f(gn)\,dn$$

is a convolution $*$-homomorphism. It is compatible with the left regular representations: $\lambda_{G/N}(Qf)$ is obtained by compressing $\lambda_G(f)$ to the $N$-invariant subspace of $L^2(G)$, identified with $L^2(G/N)$. Thus $\|\lambda_{G/N}(Qf)\| \le \|\lambda_G(f)\|$, and $Q$ extends to a surjective $*$-homomorphism

$$C_r^*(G) \longrightarrow C_r^*(G/N).$$

Surjectivity follows already on $C_c$: if $\varphi \in C_c(G/N)$, then $\varphi \circ q \in C_c(G)$ and $Q(\varphi \circ q) = \varphi$. The adjoint is a weak*-continuous isometry $B_\rho(G/N) \to B_\rho(G)$, so Lemma 1 applies again. $\square$

We also need the following standard topological reduction.

**Lemma 3.** *Every noncompact locally compact group $G$ has a noncompact second countable locally compact quotient of an open subgroup. More precisely, there are an open subgroup $H \leq G$ and a compact normal subgroup $N \lhd H$ such that $H$ is noncompact and $\sigma$-compact, while $H/N$ is second countable and noncompact.*

*Proof.* Choose a symmetric relatively compact open neighbourhood $U$ of the identity, and let $H_0$ be the subgroup generated by $U$. Then $H_0$ is open and $\sigma$-compact. If $H_0$ is noncompact, set $H = H_0$. If $H_0$ is compact, then $G/H_0$ must be infinite; otherwise $G$ would be a finite union of compact cosets. Choose representatives $g_1, g_2, \ldots$ of distinct cosets and let $H$ be the subgroup generated by $H_0 \cup \{g_1, g_2, \ldots\}$. This $H$ is open, since it contains $H_0$; it is $\sigma$-compact, because it is a countable union of finite products involving the compact set $H_0$ and finitely many of the $g_i$'s; and it is noncompact, because its image in the discrete space $G/H_0$ is infinite.

Thus we have a noncompact $\sigma$-compact open subgroup $H$. The Kakutani–Kodaira theorem [2, Theorem 8.7] gives a compact normal subgroup $N \lhd H$ such that $H/N$ is metrizable. Since $H$ is $\sigma$-compact, the locally compact metrizable group $H/N$ is second countable. Finally, $H/N$ is noncompact: if it were compact, finitely many translates of a compact identity neighbourhood in $H$, multiplied by the compact subgroup $N$, would cover $H$, forcing $H$ itself compact. $\square$

We now prove Theorem 1. Suppose, toward a contradiction, that $G$ is noncompact and $B_\rho(G)$ has weak* fpp. By Lemma 3, choose $H \leq G$ open and $N \lhd H$ compact with $H/N$ second countable and noncompact. Lemma 2(1) gives weak* fpp for $B_\rho(H)$, and Lemma 2(2) then gives weak* fpp for $B_\rho(H/N)$. Applying Naderi's Theorem 3.1 [1, Theorem 3.1] to the second countable locally compact group $H/N$, we conclude that $H/N$ is compact. This contradicts Lemma 3. Therefore $G$ must be compact.

## Verification and novelty notes

The proof is non-computational. The decisive checks are:

- the open-subgroup map is a contractive projection $C_r^*(G) \to C_r^*(H)$, hence gives a weak*-continuous isometric embedding $B_\rho(H) \hookrightarrow B_\rho(G)$;
- compact normal quotients give a contractive surjection $C_r^*(H) \to C_r^*(H/N)$, hence a weak*-continuous isometric embedding $B_\rho(H/N) \hookrightarrow B_\rho(H)$;
- weak* fpp passes to such isometric weak*-embedded dual subspaces by Lemma 1;
- the topological reduction preserves noncompactness.

Bounded novelty search was run on 2026-07-19. Cheap run indexes were searched for arXiv id `1612.08286`, the title keywords, reduced Fourier–Stieltjes algebra, weak* fpp, Bruck, and second countability, with no exact packet or attempt for this question. Web searches using the same phrases, together with $B_\rho(G)$ and Kakutani–Kodaira, found the source paper, the Fendler–Leinert separable C*-algebra paper, and Miao's reduced-dual background, but no later article explicitly removing the second countability assumption.

## Limitations and human-review recommendation

This packet answers Question 5 only. It does not address the broader BGC Questions 1–4 on page 9 of the source paper.

Human review should focus on the two functoriality steps for reduced group C*-algebras. The open-subgroup compression and compact-normal quotient maps are standard, but they are exactly where a hidden topological convention would matter.

## References


[1] F. Naderi, *C*-algebraic approach to fixed point theory generalizes Baggett's theorem to groups with discrete reduced duals*, arXiv:1612.08286; J. Math. Anal. Appl. 461 (2018), 451–460.

[2] E. Hewitt and K. A. Ross, *Abstract Harmonic Analysis. Vol. I*, 2nd ed., Springer, 1979.